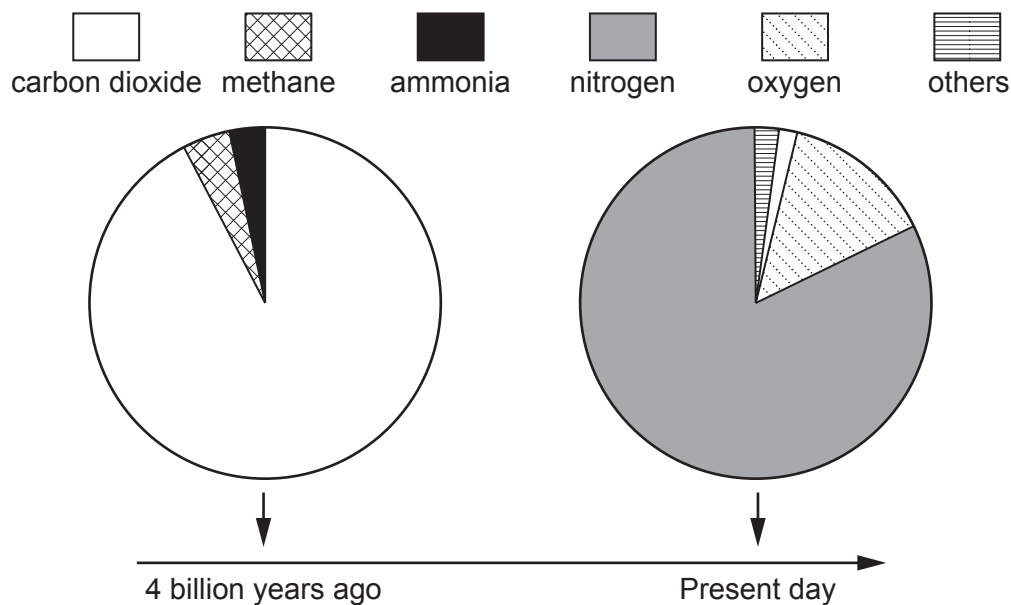


Answer **all** questions.

1. The pie charts show the composition of the air on the Earth 4 billion years ago and the present day.



- (a) (i) Give **two** reasons why the amount of carbon dioxide has changed so much over this time period. [2]
1.
2.
- (ii) State **three** other changes shown by the pie charts. [3]
1.
2.
3.



(b) The table below shows some of the other gases in the air during the present day.

Gas	Chemical formula	Approximate percentage composition (%)
argon	Ar	1.0
hydrogen	H ₂	0.00005
methane	CH ₄	0.0002
neon	Ne	0.002
nitrous oxide	N ₂ O	0.00003
ozone	O ₃	0.000004

Tick (✓) **three** correct statements about the gases in the air.

[3]

Nitrous oxide contains carbon atoms

☐

Ozone only contains oxygen atoms

☐

Hydrogen is an element that occurs as a molecule

☐

There is a higher percentage of nitrous oxide than methane

☐

The nitrous oxide molecule contains two oxygen atoms

☐

Methane is a hydrocarbon

☐
